Recommendations for Future Research

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Although the present study demonstrates that quantum layers can modify the learning dynamics of physics-informed neural networks, several research directions remain open. Future investigations should focus on improving scalability, interpretability, and practical applicability of QAPINNs.

1 Investigation on Larger and More Complex PDE Systems

The current experiments were performed on one-dimensional benchmark equations: the Allen–Cahn and Heat equations. Future work should extend the methodology to higher-dimensional and application-oriented problems, including computational fluid dynamics, turbulent flow modeling, reaction-diffusion systems, and materials-related partial differential equations. Such problems may better reveal whether quantum feature mappings provide advantages for high-dimensional solution spaces.

2 Advanced Quantum Circuit Design

The present work uses data re-uploading with ring entanglement and Pauli-Z expectation measurements. Future research should systematically compare alternative quantum architectures, including:

- problem-inspired quantum circuits,
- adaptive circuit depth optimization,
- different entanglement topologies,
- amplitude and basis encoding strategies,
- alternative measurement operators.

Automated circuit architecture search could provide a mechanism for selecting quantum circuits according to PDE characteristics rather than manually chosen architectures.

3 More Rigorous Expressivity and Trainability Analysis

The current study evaluates quantum effects through numerical performance, parameter efficiency, and gradient behavior. Future work should incorporate more formal quantum learning theory metrics, including:

- Quantum Neural Tangent Kernel (QNTK) analysis,
- Fourier spectrum decomposition of quantum feature maps,
- effective dimension measurements,
- information-theoretic analysis of learned quantum features.

These approaches could provide deeper mathematical explanations of why certain quantum circuits perform better for specific differential equations.

4 Improved Optimization Strategies

The observed sensitivity to circuit depth suggests that optimization remains a critical challenge. Future studies should investigate:

- adaptive learning rate methods,
- layer-wise quantum circuit training,
- parameter initialization strategies,
- hybrid classical-quantum optimization algorithms,
- gradient normalization techniques.

These methods may reduce training instability and mitigate potential barren plateau effects for larger quantum circuits.

5 Noisy Quantum Hardware Implementation

The current results are obtained using ideal quantum simulators. Future work should evaluate QAPINNs on real quantum processors and investigate the effects of:

- gate errors,
- measurement noise,
- decoherence,
- error mitigation techniques.

This step is necessary to determine whether the observed benefits remain valid in realistic quantum computing environments.

6 Explainable Quantum Feature Analysis

A promising direction is the development of dedicated explainability tools for quantum-enhanced neural networks. Future research could visualize and interpret:

- quantum latent representations,
- evolution of quantum expectation values during training,
- sensitivity of PDE predictions to individual quantum parameters,
- relationships between quantum features and physical quantities.

Such analyses would improve understanding of how quantum layers contribute to scientific machine learning models.

7 Towards Adaptive QAPINN Architectures

The results of this study indicate that the optimal quantum configuration is problem dependent. Therefore, future QAPINN frameworks should incorporate adaptive mechanisms that automatically determine:

- required number of qubits,
- optimal circuit depth,
- suitable encoding method,
- appropriate measurement strategy.

An adaptive architecture could reduce computational cost while maintaining the benefits of quantum feature representations.